

Assignment 3

A- Nested Loops Tasks

1. Right Triangle Number Pattern

Write a program to print a right triangle number pattern using nested loops.

Example:

Input: 5

Output:

1

22

333

4444

55555

2. Hollow Right Triangle Star Pattern

Write a program to print a hollow right triangle star pattern using nested loops.

Example:

Input (number of rows): 6

Output:

*

* *

* *

* *

* *

* * * * *

3. Inverted Right Triangle Star Pattern

Write a program to print an inverted right triangle star pattern using nested loops.

Example:

Input (number of rows): 6

Output:

**

*

4. List of Non-Prime Numbers

Write a program to take a number as input and print a list of non-prime numbers from 1 to the given number using nested loops.

Example:

Input: 25

Output: 1 4 6 8 9 10 12 14 15 16 18 20 21 22 24 25

5. Number to Words Conversion (Note: You don't need nested loops here)

Write a program to input any number and print it as words **from left to right**.

Example:

(Note: The input should be one number not a number for each digit)

Input: 8309

Output: Eight Three Zero Nine

B- Array Tasks

(NOTE: Use the arrays given in each question)

6. Count Even and Odd Elements in an Array

Write a program to input elements in an array from the user and count how many of the elements are even and how many are odd.

Example:

- **Input** array: 1 2 3 4 5 6 7 8 9

- **Output:**

Total even elements: 4

Total odd elements: 5

7. Search for an Element in an Array

Write a program to input elements into an array from the user and search whether a given element exists in the array. If the element exists, display its index **(You don't have to find all the occurrences of the given element. The first occurrence is enough)**.

Example:

- **Input** array: 10 12 20 25 13 10 9 40 60 5

Element to search: 25

- **Output:**

Element found at index 3

8. Count the Number of Occurrences of an Element in an Array

Write a program to input elements into an array from the user and count the number of occurrences of a specified element.

Example:

- **Input** array: 5 7 8 8 5 8 7 7

Element to count: 7

- **Output:** Number of occurrences of 7: 3

9. Copy Elements from One Array to Another

Write a program to input elements into an array from the user and copy all elements from the first array into a second array.

Example:

- **Input** array1: 10 1 95 30 45 12 60 89 40 -4

- **Output:**

Array1: 10 1 95 30 45 12 60 89 40 -4

Array2: 10 1 95 30 45 12 60 89 40 -4

10. Find the Additive Inverse of an Array

Given an array of integers, print a new array where each value is replaced by its additive inverse (**positive values become negative, and negative values become positive**).

Examples:

- **Input:** 1 2 3 4 5

Output: -1 -2 -3 -4 -5

- **Input:** 1 -2 3 -4 5

Output: -1 2 -3 4 -5

11. Double the Elements of an Array

Given an array of integers, print a new array where each element is doubled.

Examples:

- **Input:** 1 2 3

Output: 2 4 6

12. Find Unique Elements in an Array

Write a program to input elements into an array from the user and print all unique elements (**elements that appear only once**).

Example:

- **Input** array: 1 2 3 5 1 5 20 2 12 10

- **Output:** All unique elements in the array are: 3 20 12 10

13. Find Pairs of Elements Whose Sum Equals The Last Element

Write a program to input elements into an array from the user and find all pairs of numbers whose sum equals the last element in the array.

Example:

- **Input** array: 1 5 7 5 8 9 11 12

- **Output:**

- Array pairs whose sum equals 12:

1 11

5 7

7 5

14. Determine if the Sum of Array Elements is Odd or Even

Write a program to determine if the sum of an array's elements is odd or even. You should allocate an array with a fixed size of 10 elements. Ask the user how many elements they want to enter (up to 10). Then, take input from the user for those elements, calculate the sum of the entered numbers, and print "odd" or "even" as the result. **(Note: if the user entered 0 for the number of elements, the output should be "even")**

Examples:

- **Input:** number of elements: 0

Output: even

- **Input:** number of elements: 3

Array elements: 0 1 4

Output: odd

- **Input:** number of elements: 3

Array elements: 0 -1 -5

Output: even

15. Find the Element Without a Matching Pair

You will be given an array of integers where every element has both a positive and negative counterpart, except for one element. Your task is to find the element that has no matching positive or negative value.

Examples:

- **Input:** 1 -1 2 -2 3 4 -4

Output: 3

(3 has no matching negative appearance)

- **Input:** -3 1 2 3 -1 -4 -2

Output: -4

(-4 has no matching positive appearance)

- **Input:** 1 -1 3 2 -2 3 3

Output: 3

(Note: The only positive or only negative integer may appear more than once like number 3 here.)

16. Delete an Element from an Array

Write a program to delete an element from an array at a specified position.

Example:

- **Input** array: 10 20 30 40 50

Position to delete: 2

- **Output:** Array elements: 10 30 40 50

BONUS PROBLEMS

1. Pyramid Star Pattern

Write a program to print an equilateral triangle or pyramid star pattern series with `n` rows using a loop. Example:

Input: 5

Output:

```
  *
 ***
*****
*****
*****
```

2. Rhombus Star Pattern

Write a program to print a rhombus star pattern series with `n` rows using a loop.

Example:

Input: 6

Output:

```
  *****
 *****
 *****
 *****
 *****
 *****
```

3. Codeforces Problem - Count Words

Solve the following Codeforces problem: [Problem W - Count Words]

(<https://codeforces.com/group/MWSDmqGsZm/contest/219432/problem/W>).

4. Codeforces Problem - Sum Digits

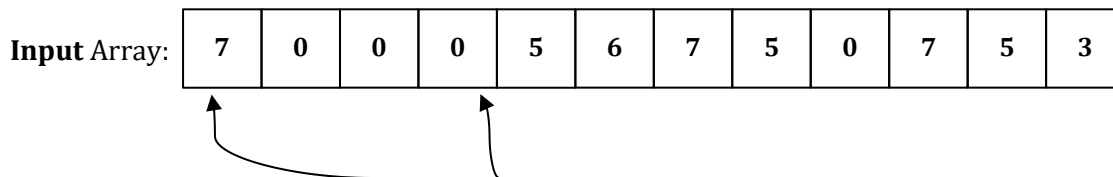
Solve the following Codeforces problem: [Problem U - Sum Digits]

(<https://codeforces.com/group/MWSDmqGsZm/contest/219432/problem/U>).

5. Find the Longest Distance Between Two Equal Elements in an Array

Write a program to find the longest distance between two identical elements in an array of integers. The distance is measured by the number of cells between the two elements.

Example:



- The distance between the first and fourth cells (values: 7 and 0) is 2 (cells 2 and 3 are in between).
- The longest distance in the example is between the first 7 (index 0) and the tenth 7 (index 9), with a distance of 8 cells.

Note:

1. The array values will be provided by the user.
2. If all input elements in the array are identical (e.g., [1, 1, 1, 1, 1]), the distance is calculated as the number of cells between the first and the last element.